If there are 'x' pens and the cost of each pen is ₹ 'y', then the cost of $(3x^2)$ pens bought by Geetha @ ₹ $5y$

@ - at the rate of

$$= (3x^2) \times 5y$$
$$= (3 \times 5)(x^2 \times y)$$
$$= ₹\ 15x^2y$$

Example 3.1

If the length and breadth of a rectangular painting are $4xy^3$ and $3x^2y$. Find its area.



Solution:

Area of the rectangular painting, $A = (l \times b)$ *sq.units*

$$= (4xy^3) \times (3x^2y)$$
$$= (4 \times 3)(x \times x^2)(y^3 \times y)$$
$$A = 12x^3y^4 \text{ sq.units}$$

Example 3.2

Find the product of $2x^2y^2$, $3y^2z$ and $-z^2x^3$

Solution:

We have, $(2x^2y^2) \times (3y^2z) \times (-z^2x^3)$

$$= (+) \times (+) \times (-)(2 \times 3 \times 1)(x^2 \times x^3)(y^2 \times y^2)(z \times z^2)$$
$$= -6x^5y^4z^3$$



Try these

Find the product of

(i) $3ab^2, -2a^2b^3$

(ii) $4xy, 5y^2x, (-x^2)$

(iii) $2m, -5n, -3p$

3.2.2 Multiplication of a polynomial by a monomial

If there are 'a' shops and each shop has 'x' apples in 8 baskets and 'y' oranges in 3 baskets and 'z' bananas in 5 baskets, then the total number of apples, oranges and bananas are

$= a \times (8x + 3y + 5z)$

$= a(8x) + a(3y) + a(5z)$

(using distributive law).

$= 8ax + 3ay + 5az$



monomial × monomial	= monomial
binomial × monomial	= binomial
binomial × binomial	= binomial/polynomial
polynomial × monomial	= polynomial



Note

Distributive law	If a is a constant, x and y are variables then $a(x + y) = ax + ay$ For example, $5(x + y) = 5x + 5y$

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